

5 (long) A set of atoms $N(t)$ decay into “daughter” atoms at a rate given by $\lambda N(t)$ so that

$$\frac{dN(t)}{dt} = -\lambda N(t)$$

where λ is a positive constant. We assume that $N = N_0$ at $t = 0$.

(a) Find an expression for $N(t)$, and show that after a time $T_{1/2} = \ln(2)/\lambda$ the number of the original atoms left has halved. [5]

(b) If the daughter atoms N_1 do not themselves decay, derive an expression for $N_1(t)$ assuming that $N_1(t) = 0$ at $t = 0$. [5]

(c) Suppose instead that the daughter atoms, N_1 , also decay with a decay constant given by λ^* producing their own daughter atoms, N_2 , which we assume do not decay. Show this means that

$$\frac{dN_1(t)}{dt} + \lambda^* N_1(t) = \lambda N_0 \exp(-\lambda t)$$

and, finding the particular integral for this equation for $N_1(t)$, solve for $N_1(t)$ under these new conditions, assuming again that $N_1(t) = 0$ at $t = 0$. [13]

(d) At $t = 0$, $N_1 = 0$, and as $t \rightarrow \infty$, $N_1(t) \rightarrow 0$. $N_1(t)$ is positive for $t > 0$. Derive an equation for the time, t^* , when the number of atoms $N_1(t^*)$ is a maximum. Show that t^* is positive where $\lambda > \lambda^*$ or $\lambda < \lambda^*$. [7]